How to simulate data

review of some simulation settings in causal inference

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My settings

N obs data, M rct data; d covariates, within which c is confounder
[center the response?]
1. N/M
2. $e(x)$, overlap, well-specified or not
3. c/d, how many irrelevant covariates
4. d(c)/N, high-dimensional settings or not
5. sample size, how much information

Simulation settings

(Li, Thomas, and Li 2019) (Yang, Thomas, and Li 2026) (how much overlap, i.e. how strong tails of $e(x)$)

- X

$$V = (V_1, \ldots, V_6)^\top \sim \mathcal{N}(0, \Sigma_{6\times 6}), \quad \Sigma_{jj} = 1, \quad \Sigma_{jk} = 0.5 \ \ (j \neq k),$$

$$X_j = \begin{cases} V_j, & j = 1, 2, 3 (\text{continuous}), \\ \mathbf{1}_{\{V_j < 0\}}, & j = 4, 5, 6 (\text{binary}). \end{cases}$$

- $Z \mid X$ (vary γ from 1 to 4 to reflect increasingly sparse overlap between groups)

$$e(X) = \{1 + \exp[-(\alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 + \alpha_4 X_4 + \alpha_5 X_5 + \alpha_6 X_6)]\}^{-1}$$

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6) = \gamma \times (0.15, 0.3, 0.3, -0.2, -0.25, -0.25)$$

- $Y \mid X$ (a homogenous and additive treatment effect Δ)

$$E(Y|Z, X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \Delta Z$$

$$\beta = (0, -0.5, -0.8, -1.2, 0.8, 0.8, 1)$$

(so that subjects with lower outcomes are more likely to receive the treatment)

$$Y \sim N(E(Y \mid X), 1.5)$$

- model misspecification

i) wrong functional form (mistaking X_3 by X_3^2) and ii) omitting variables (X_3 and X_4).

(Wager and Athey 2018) (how $\tau(x)$ related with X, HTE)
Main effect: $m(x) = 2^{-1}\mathbb{E}[Y^{(0)} + Y^{(1)} \mid X = x]$

- X

$$X \sim U([0,1]^d)$$

- $Y \mid X$, $Z \mid X$

$$Y^{(0/1)} \sim N(E[Y^{(0/1)} \mid X], 1)$$

  - non-random assignment $e(X) = \frac{1}{4}(1 + \beta_{2,4}(X_1))$, non-zero main effect, homogeneous

$$m(X) = 2X_1 - 1, \quad \tau(X) = 0$$

  - random assignment $e(X) = 0.5$, zero mean effect $m(X) = 0$, heterogeneous

$$\tau(x) = \varsigma(X_1)\varsigma(X_2), \quad \varsigma(x) = 1 + \frac{1}{1 + e^{-20(x-1/3)}}$$

  - random assignment $e(X) = 0.5$, zero mean effect $m(X) = 0$, heterogeneous[1]

$$\tau(x) = \varsigma(X_1)\varsigma(X_2), \quad \varsigma(x) = \frac{2}{1 + e^{-12(x-1/2)}}$$

- evaluation: We evaluate performance in terms of expected mean-squared error for estimating $\tau(X)$ at a random test example X, as well as expected coverage of $\tau(X)$ with a target coverage rate of 0.95.

[1] this setting for: weakness of nearest neighbor approaches(random forests) they can fill the valleys and flatten the peaks of the true $\tau(x)$ function, especially near the edge of feature space

(Wager 2024) [HTE, compare lasso-based R-learner and T-learner]

- X

$$X \sim \mathcal{N}\left(0, I_{10\times 10}\right)$$

- $Z \mid X$, $Y \mid X$(regularization-induced confounding)[2]

$$e(X) = 1/\left(1 + e^{-(X_2+X_3)}\right)$$

$$Y(z) = 2\log\left(1 + e^{X_1+X_2+X_3}\right) + zI\left(X_2 + X_3 \geq 0\right) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,1)$$

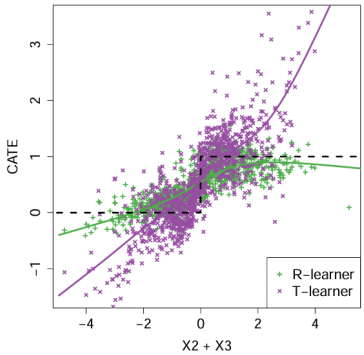

Figure 1: regularization-induced-confounding

[2]this setting for: units for which $X_2 + X_3$ is large are simultaneously more likely to be treated, have a larger baseline effect whether or not they get treated, and have a larger treatment effect.

(Diamond and Sekhon 2013) (Lee, Lessler, and Stuart 2010)

Setting 1

- X ten covariates $(X_k, k = 1, ..., 10)$: four confounders associated with both treatment and outcome, three treatment predictors(instruments[3]), and three outcome predictors. Six covariates $(X_1, X_3, X_5, X_6, X_8, X_9)$ are binary, whereas four (X_2, X_4, X_7, X_{10}) are standard normal.

- $Z \mid X$ (change the degree of linearity and additivity in the true propensity score model) (the average probability of treatment assignment at the average value of the covariates is ≈ 0.5)

  A: Additivity and linearity (mean effects only)

  B: Mild nonlinearity (one quadratic term)

  C: Moderate nonlinearity (three quadratic terms)

  D: Mild nonadditivity (three two-way interaction terms)

  E: Mild nonadditivity and nonlinearity (three two-way interaction terms and one quadratic term)

  F: Moderate nonadditivity (ten two-way interaction terms)

  G: Moderate nonadditivity and nonlinearity (ten two-way interaction terms and three quadratic terms)

- $Y \mid X$ (homogeneous effect, linear model)

$$Y = \gamma Z - 3.85 + 0.3X_1 - 0.36X_2 - 0.73X_3 - 0.2X_4 + 0.71X_8 - 0.19X_9 + 0.26X_{10}$$

Setting 2

- X based on the LaLonde(1986) data. Some of the baseline variables are discrete, and others contain pointmasses and skewed distributions. None of the covariates,as they are based on real data, have ellipsoidal distributions[?].

- $Z \mid X$ (not correctly specified propensity model)

$$\begin{aligned}\pi_i = \text{logit}^{-1}\big[&1 + .5\hat{\mu} + .01\,\text{age}^2 - .3\,\text{educ}^2 \\ &- .01\log(\text{re74} + .01)^2 + .01\log(\text{re75} + .01)^2\big],\end{aligned} \tag{4}$$

  This propensity score is a mix of the estimated propensity score plus extra variables, so that the propensity model estimated in the Monte Carlo samples is badly misspecified[4].

- $Y \mid X$

$$\begin{aligned}Y =& 1000\,T + .1\exp\big[.7\log(\text{re74} + .01) \\ &+ .7\log(\text{re75} + 0.01)\big] + \epsilon,\end{aligned}$$

[3] In practice, instruments should not be included in the propensity score when assuming selection on observed variables (and not in an OLS regression model if that is used)[?]

[4] GenMatch is able to obtain better levels of balance without requiring the analyst to correctly specify the propensity score.

(Rubin 1973) [ATT, homogeneous/parallel, matching]

- $X \mid Z = z$

$$X \mid Z = 1 \sim \mathcal{N}(\eta_1, \sigma_1^2), \quad X \mid Z = 0 \sim \mathcal{N}(\eta_2, \sigma_2^2)$$

  where

$$\eta_1 = -\eta_2, \ (\sigma_1^2 + \sigma_1^2)/2 = 1$$

  - variance ratio: $\sigma_1^2/\sigma_2^2 \in \{1/2, 1, 2\}$ (a ratio of 1/2 is more "favorable to match" than 1 because the target group is more concentrated than the reservoir group).
  - mean discrepancy: $B = 2\eta_1 \in \{1/4, 1/2, 3/4, 1\}$, which is the standard deviation between the means of X

- $Y(z) \mid X$, response surface $V(x)$ (Robustness to poor overlap)

$$Y_z(x) = \alpha_z + V(x), \ V(0) = 0, \ \alpha_1 - \alpha_0 = \tau$$

$$V(x) = exp(ax), \ a \in \{1, 1/2, -1/2, -1\}$$

  Since $\eta_1 > \eta_2$, compared with $exp(x/2)$, the response surfaces $exp(x)$ is "less robust to poor overlap" because it is increasing most rapidly for large X which is where close matches are difficult to find, thus this is "less favorable to matching" than $exp(x/2)$.

- sample size of treatment N and control groups rN

$$r \in \{2, 3, 4\}$$

(D'Amour et al. 2021) (strict overlap and gaussian covariates)

- If X is one-dimensional and follows a Gaussian distribution under both P_0 and P_1, **strict overlap implies that $P_0 = P_1$, or that the covariate is perfectly balanced**. Similar results can be derived when $X_{1:p}$ is multi-dimensional Gaussian. Thus, for Gaussianly distributed covariates, the implications of strict overlap are so strong that they are uninteresting.

- simulation intuition
  - $P_1 = P_0 \sim N(\mu, \Sigma)$
  - $P_1 \neq P_0,\ P_1 \sim N(\mu_1, \Sigma_1),\ P_0 \sim N(\mu_0, \Sigma_0)$
  - Truncated Normal
  - Uniform/Beta/..

(Shi et al. 2025) [simple, but code available]

2.3. Simulation Study

In this subsection, we conduct simulation studies using synthetic data from a PLR model to compare the finite-sample performance of the DML estimator with some benchmark estimators. We consider a scenario where the outcome variable Y and the treatment variable D both depend nonlinearly on two covariates $\{X_1, X_2\}$ that serve as confounders. Specifically, D is drawn from a logistic regression model, and Y is drawn from a second-order polynomial regression with an additive treatment effect. Note that we simulate a binary treatment D only as an example, but D can be also a multi-valued or continuous variable. The data generation process is as follows:

$$X_1 \sim N(0, 2^2),\ X_2 \sim N(0, 2^2),\ \Pr(D = 1|X_1, X_2) = \frac{1}{1 + e^{-(X_1 + X_2 + 1)}},$$
$$Y = 1 + D + X_1 + X_2 + X_1 X_2 + X_1^2 + X_2^2 + \varepsilon,\ \varepsilon \sim N(0, 1^2),\ \varepsilon \perp (X_1, X_2).$$

We are interested in estimating the coefficient $\theta_0 = 1$. We generate synthetic data for varying sample sizes $N \in \{100, 500, 1000, 5000, 10000\}$. Note that the data generation process above is unknown to researchers in practice – researchers only observe a particular sample $(Y_i, D_i, X_{1i}, X_{2i})_{i=1}^N$ and have to decide on model specification according to contexts, theories, or empirical conventions. Under

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